Software Requirements Specification

for

Loan Eligibility Prediction System (LEPS)

Version 1.0

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Revision History

Name	Date	Reason for Changes	Version
Student / Team	11 Sep 2026	Initial SRS prepared from the LEPS feasibility report	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for the Loan Eligibility Prediction System (LEPS), Version 1.0. LEPS is a web-based system for assessing loan eligibility from applicant, financial, employment and credit-bureau data. It produces a calibrated eligibility probability, risk tier and model-derived explanation while retaining the final decision with an authorized loan officer.

1.2 Document Conventions

The document follows the IEEE-style SRS structure supplied in the lab template. Requirements use unique REQ identifiers. “Shall” denotes a mandatory requirement. Priority is High, Medium or Low where stated. TBD is used only where the source feasibility report does not define sufficient detail.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for developers, testers, project managers, loan officers, risk/policy personnel, compliance/audit personnel, system administrators and academic evaluators. Readers should begin with Sections 1 and 2, then use Sections 3–5 for interfaces, functional requirements and quality constraints. Administrators and compliance stakeholders should also review the security, audit and business-rule sections.

1.4 Product Scope

LEPS collects and validates structured applicant information, generates an ML-based eligibility prediction, presents probability and risk tier, explains the principal factors affecting the prediction, and supports officer approval, decline or escalation. It records predictions and officer actions in an immutable audit trail and supports monitoring of model performance and prediction-distribution drift. Actual loan disbursal, servicing, collections and fully autonomous final decisions are outside scope.

1.5 References

Software Engineering & Design Principles Lab, Experiment 2 Report: Identification of Problem, Objectives, Scope and Feasibility Analysis; Project: Loan Eligibility Prediction System (LEPS). Source: supplied feasibility report.

2. Overall Description

2.1 Product Perspective

LEPS is a new web application intended to augment the existing loan review workflow. The current workflow consists of application intake, credit-bureau pull, scorecard/policy evaluation, manual review and decision/disclosure. LEPS automates the scoring and explanation step while retaining human review for the final decision.

2.2 Product Functions

• Authenticate users and enforce role-based access control.

• Capture applicant, employment, income, debt, credit and loan-request data.

• Validate data for completeness, format and acceptable ranges.

• Generate a calibrated eligibility probability and risk tier using a served ML model.

• Provide a ranked, plain-language explanation of factors supporting or reducing eligibility.

• Allow authorized officers to approve, decline or escalate applications and record overrides with notes.

• Maintain a reconstructable audit record containing prediction inputs, model version, output, explanation and officer actions.

• Monitor model performance and prediction-distribution drift over time.

2.3 User Classes and Characteristics

User Class	Characteristics / Responsibilities
Applicant / Borrower	Provides application details and receives the eligibility outcome and guidance where applicable.

Loan Officer / Underwriter	Reviews the application, prediction and explanation; confirms or overrides the recommendation.
Risk / Credit Policy Team	Reviews portfolio-level outcomes and supports policy/model calibration.
Compliance / Audit Team	Verifies compliance and reconstructs the rationale for decisions.
System Administrator	Manages users, roles and model-version promotion.

2.4 Operating Environment

• Frontend: React with TypeScript and Tailwind CSS.

• Backend: Python with FastAPI.

• Database: PostgreSQL 15+.

• ML: scikit-learn with XGBoost or LightGBM; SHAP for explainability; MLflow for model registry.

• Deployment/development: Docker, Docker Compose, Git and GitHub Actions.

• Development may run on Windows, macOS or Linux; deployment is planned in Linux containers.

2.5 Design and Implementation Constraints

• The proposed React/TypeScript, FastAPI and PostgreSQL stack shall be used unless an approved project change is made.

• The ML solution shall support CPU-based pilot deployment; no GPU is required.

• The final decision shall remain subject to authorized human review.

• Prediction records must retain sufficient information to reconstruct the decision rationale.

• Production deployment must consider jurisdiction-specific compliance requirements.

2.6 User Documentation

The system should include concise guidance for applicants and loan officers covering application entry, interpretation of probability/risk outputs and officer overrides. Administrator guidance should cover user/role management and model-version promotion. Exact delivery format is TBD.

2.7 Assumptions and Dependencies

• Representative training data is available and suitable for the intended lending context.

• Credit-bureau information is available in the required structured or summarized form.

• The selected ML model and model registry can be served in the deployment environment.

• Users have access to a modern web browser and network connectivity.

• Regulatory and compliance requirements may vary by jurisdiction and must be reviewed before production use.

3. External Interface Requirements

3.1 User Interfaces

• The applicant interface shall provide structured forms for personal, employment, income, debt, credit and loan-request information.

• Validation errors shall be clearly associated with missing or invalid fields.

• The prediction view shall present eligibility probability, risk tier and ranked explanation.

• The officer dashboard shall present the application, prediction and explanation together.

• Officer actions shall include approve, decline and escalate; an override note is required when the recommendation is changed.

• Authorized compliance/audit users shall have an interface for retrieving relevant decision records.

3.2 Hardware Interfaces

No specialized hardware interface is required. Pilot deployment is expected to use approximately 2–4 vCPU, 4–8 GB RAM and 20–50 GB SSD storage. No GPU is required.

3.3 Software Interfaces

• React/TypeScript communicates with FastAPI backend services.

• FastAPI communicates with PostgreSQL for application, prediction, decision and audit data.

• The backend invokes the served ML model for eligibility prediction.

• SHAP provides model-derived explanation factors.

• MLflow provides model registry and version-management functionality.

• Docker/container tooling packages the application and supporting services.

3.4 Communications Interfaces

The system shall use standard web communication between browser, backend API and database services. HTTPS/TLS should be used in deployed environments. Authentication information shall be transmitted securely. Exact API message schemas are implementation details and are TBD.

4. System Features

4.1 Authentication and Role-Based Access

Description and Priority: High. The system shall authenticate users and restrict functions according to their assigned role.

Stimulus/Response Sequences

• Valid credentials submitted → system authenticates the user and opens the permitted dashboard.

• Restricted operation attempted → system denies access and displays an authorization error.

• Administrator updates a role → subsequent access reflects the authorized permissions.

Functional Requirements

• REQ-1: The system shall authenticate registered users before granting access to protected functions.

• REQ-2: The system shall enforce role-based access for officer, risk/policy, compliance/audit and administrator functions.

• REQ-3: The system shall deny unauthorized access to protected applicant, prediction and audit information.

• REQ-4: The system shall allow administrators to manage user access and roles.

4.2 Applicant Data Intake and Validation

Description and Priority: High. The system shall collect the structured data required for eligibility assessment and validate it before prediction.

Stimulus/Response Sequences

• Application data entered → system checks completeness, format and ranges.

• Invalid or missing information detected → system identifies the affected fields and prompts for correction.

• Valid application submitted → system permits prediction processing.

Functional Requirements

• REQ-5: The system shall capture age, address and number of dependents.

• REQ-6: The system shall capture employment type, employment tenure and declared income.

• REQ-7: The system shall capture existing EMIs, credit-card outstanding and information needed to derive debt-to-income ratio.

• REQ-8: The system shall capture bureau score and summarized credit-report fields such as delinquency history and credit age.

• REQ-9: The system shall capture requested loan amount, tenor and purpose.

• REQ-10: The system shall prevent prediction when mandatory fields are missing or invalid.

• REQ-11: The system shall provide actionable validation messages for incomplete, incorrectly formatted or out-of-range data.

4.3 Eligibility Prediction and Risk Tier

Description and Priority: High. The system shall use a served ML model to produce a calibrated eligibility probability and risk tier.

Stimulus/Response Sequences

• Valid application submitted → backend sends validated features to the prediction service.

• Prediction succeeds → probability and risk tier are displayed and recorded.

• Prediction fails → system shall report failure and shall not fabricate an eligibility result.

Functional Requirements

• REQ-12: The system shall return an eligibility probability for a valid application.

• REQ-13: The system shall assign a risk tier associated with the prediction.

• REQ-14: The system shall associate each prediction with the model version used.

• REQ-15: The system shall record prediction inputs, output and explanation information for audit purposes.

4.4 Prediction Explainability

Description and Priority: High. The system shall explain predictions using model-derived factors in plain language.

Stimulus/Response Sequences

• Prediction generated → explanation mechanism derives principal contributing factors.

• User views prediction → ranked positive and negative factors are displayed.

• Declined/borderline application → improvement factors are shown where determinable from the model.

Functional Requirements

• REQ-16: The system shall provide a ranked explanation for each prediction.

• REQ-17: The explanation shall identify factors that positively and negatively influence eligibility.

• REQ-18: The explanation shall be expressed in plain language.

• REQ-19: The explanation shall be derived from the underlying prediction model rather than a generic message.

• REQ-20: For declined or borderline applications, the system shall identify factors that could most improve eligibility where determinable.

4.5 Officer Decision and Override

Description and Priority: High. The system shall support human-in-the-loop review and final officer action.

Stimulus/Response Sequences

• Officer opens application → application, prediction and explanation are displayed.

• Officer confirms recommendation → decision is recorded.

• Officer changes recommendation → system requires an override note and records the action.

Functional Requirements

• REQ-21: The system shall allow an authorized officer to approve, decline or escalate an application.

• REQ-22: The system shall require a note when an officer overrides the recommendation.

• REQ-23: The system shall record officer identity, action and decision time.

• REQ-24: The system shall retain the original model recommendation when an officer overrides it.

4.6 Audit Trail

Description and Priority: High. The system shall maintain an immutable and queryable record of predictions and decisions.

Stimulus/Response Sequences

• Prediction generated → audit record is created.

• Officer acts → action is appended to audit history.

• Authorized compliance user requests a record → reconstructable decision history is returned.

Functional Requirements

• REQ-25: The system shall record prediction inputs, model version, output and explanation.

• REQ-26: The system shall record every officer decision and override.

• REQ-27: Audit records shall be protected against unauthorized modification.

• REQ-28: Authorized compliance/audit users shall be able to query records needed to reconstruct a decision.

4.7 Model Performance and Drift Monitoring

Description and Priority: Medium. The system shall support monitoring of model performance and prediction-distribution drift over time.

Stimulus/Response Sequences

• Monitoring data accumulates → configured monitoring metrics are updated.

• Change in performance/distribution detected → monitoring result is exposed to authorized stakeholders.

• Model version reviewed/promoted → relevant version is recorded.

Functional Requirements

• REQ-29: The system shall support tracking model performance metrics over time.

• REQ-30: The system shall support detection or reporting of prediction-distribution drift.

• REQ-31: The system shall associate monitoring results with the relevant model version where applicable.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

• The system should provide responsive form validation and dashboard navigation.

• Prediction processing should support interactive officer use in the pilot environment.

• The pilot deployment shall operate without a GPU.

• Exact response-time and throughput targets are TBD because the feasibility report does not specify quantitative thresholds.

5.2 Safety Requirements

The system shall not be treated as an autonomous lending authority. Final decisions shall remain subject to authorized human review. A failed or unavailable prediction shall never be represented as a valid eligibility result. Production use should undergo jurisdiction-specific compliance review.

5.3 Security Requirements

• Authentication shall be required for protected functions.

• Role-based authorization shall restrict access to applicant, prediction, decision and audit information.

• Sensitive applicant and credit information shall be protected against unauthorized access.

• Audit records shall be protected against unauthorized alteration.

• HTTPS/TLS should be used for deployed web communications.

• Logs shall support compliance investigation without exposing unnecessary sensitive information.

5.4 Software Quality Attributes

• Usability: application, prediction and explanation should be visible together to minimize workflow overhead.

• Reliability: prediction failures must fail safely without inventing a result.

• Maintainability: model versions and application components should be independently manageable.

• Auditability: decisions must be reconstructable from retained prediction and action records.

• Explainability: prediction reasons should be understandable and faithful to the underlying model.

• Scalability: the architecture should support increased application volume without proportional manual effort.

5.5 Business Rules

• BR-1: Only authorized roles may access assigned functions.

• BR-2: Prediction may be attempted only after required application data passes validation.

• BR-3: Model output is an assessment/recommendation and does not replace authorized human final decision-making.

• BR-4: Officer override requires a mandatory note.

• BR-5: Every prediction and officer decision must have an audit record.

• BR-6: Audit history must be reconstructable and protected from unauthorized modification.

• BR-7: Decline guidance must be based on factors supported by the prediction process rather than generic reasons.

6. Other Requirements

Database requirements: PostgreSQL 15+ shall store structured application, prediction, decision, user/role and audit information. The database shall support relationships among an application, predictions, model version and officer actions.

Legal/compliance requirements: Real-world deployment shall consider applicable lending, privacy and fair-lending requirements for the target jurisdiction. Exact regulatory controls are TBD.

Internationalization: No specific localization requirements are defined in the feasibility report; these are TBD.

Appendix A: Glossary

Term	Definition
LEPS	Loan Eligibility Prediction System.
ML	Machine Learning.
SHAP	Model-explanation approach used to identify feature contributions to predictions.
XGBoost / LightGBM	Gradient-boosting frameworks suitable for tabular prediction.
Risk Tier	Risk category associated with the calibrated eligibility prediction.
Audit Trail	Queryable record of predictions and officer actions used to reconstruct decision rationale.
Drift	Change in input or prediction distributions over time that may affect model behavior.
LOS	Loan Origination System.
NBFC	Non-Banking Financial Company.
CIBIL	Credit-bureau service referenced in the feasibility report.

Appendix B: Analysis Models

The supplied template permits data-flow, class, state-transition and entity-relationship models. The feasibility report does not contain finalized diagrams, so the following logical model is provided as the source-supported analysis description.

High-level flow: Applicant/Officer → Data Intake & Validation → ML Prediction → Probability + Risk Tier → Explanation → Officer Decision → Audit Trail.

Primary entities: User/Role, Loan Application, Applicant Data, Credit Data, Prediction, Model Version, Explanation, Officer Decision and Audit Record.

External/system components: Web Browser, FastAPI backend, PostgreSQL database, ML model service, SHAP explanation component and MLflow model registry.

Appendix C: To Be Determined List

ID	Open Item
TBD-1	Exact quantitative response-time and throughput targets.
TBD-2	Exact user-interface layouts and screen designs.
TBD-3	Exact API message schemas and endpoint specifications.
TBD-4	Exact authentication/session policy.
TBD-5	Exact jurisdiction-specific regulatory and privacy controls.
TBD-6	Exact database schema and audit-record retention periods.
TBD-7	Exact risk-tier thresholds and calibration thresholds.
TBD-8	Exact model-performance and drift thresholds.
TBD-9	Exact user-documentation delivery format.
TBD-10	Internationalization/localization requirements.